

CCP6224 Object Oriented Analysis and Design

Lab 8 – Analysis, Design & UML

Analysis and System Design

1. The following scenarios depict different levels of cohesion in a software design
 - a) Consider the following pseudocode for an object class which is used for generating monthly reports for a system

```
class Report{  
    void connectDB();  
    void generateMonthlySalesReport();  
    void getCurrentWeather();  
    void restartServer();  
}
```

- What type of *cohesion* do you think it depicts and why?
- What kinds of problems can arise with the type of *cohesion* you identified?
- How can the class be improved upon to reduce/avoid the problems you claimed?

- b) What about the following object class?

```
class Cake{  
    void addEggs(int num);  
    void addSugar(double amnt);  
    void addFat(double amnt);  
    void beatIt(double woohoo);  
}
```

2. The following interface for a `Person` object is said to break the *Interface Segregation* principle of OO design. Explain this problem and how it can be corrected

```
public interface Person{  
    public String getName();  
    public double getSalary();  
    public Date getBirthdate();  
    public Schedule getClassSchedule();  
    public Schedule getWorkSchedule();  
    public void setSalary();  
}
```

UML Structural Diagrams

1. Draw a class diagram to represent the statements

“A company hires many employees but an employee only works at one company. Each company has a name and address while employees have a name and salary”.

Then draw an object diagram for the same scenario above showing a running instance of the class diagram model.

2. You are given the following scenario for a library loan system
 - The system keep tracks of when items (books, magazines, journals, references) are borrowed and returned
 - The system must support librarian work
 - The library is open to staff and active students
 - Staff may borrow up to 20 different items for up to a month
 - Students may borrow up to 10 different items for a week
 - Librarians handle the loans from members of the library

Based on the scenario above, draw a class diagram for the system.

3. Draw a class diagram representing the relationship between parents and children. Take into account that a person can have both a parent and a child AND also BE a parent and a child. Annotate associations with roles and multiplicities
4. Draw the UML state diagram for the following scenario

A security alarm system has the following states: idle, armed and triggered. When the system is started it is and remains in idle state if no action is performed until it is armed by setting the security code. The alarm is triggered if a sensor is tripped during an intrusion and remains in alarm state if the sensor is active. The alarm stops ringing and goes back to armed if the correct security code is entered, but stays in the triggered state if the wrong code is entered.

5. What type of coupling is this?

Class `CourseSection` has public class variables called `minClassSize` and `maxClassSize`. These are changed from time to time by the university administration. Many methods in classes `Student` and `Registration` access these variables.

How can we improve this?